Govt. Of Bihar

Department of Science and Technology

GOVERNMENT ENGINEERING COLLEGE SIWAN

1st Mid Semester Examination 2025

Time: 1.5 hr

Subject: Engineering Physics

Semester: 1st

B.tech.: CSE,CSEIOT,VLSI & EE

Max. Marks: (4x5=20)

Instructions: Question **1 is compulsory** Attempt any **Four from 2-8** questions. All questions carry equal marks.

1) Choose the correct options

(i) Electric field is the: (A) Gradient of potential. (B) Negative gradient of potential (C) Curl of potential (D) Divergence of potential

(ii) Stokes' Theorem relates: (A) Volume integral to line integral (B) Surface integral of a curl to a line integral (C) Line integral to scalar potential (D) Surface integral to volume integral

(iii) The curl of a gradient of any scalar field is always: (A) Zero (B) Unity (C) Divergence (D) Infinity

(iv) The gradient of a scalar field (ϕ) represents: (A) Rate of change of (ϕ) in space (B) Total flux of (ϕ). (C) Direction of zero change of (ϕ) (D) Circulation of (ϕ)

2) Explain **intrinsic** and **extrinsic** semiconductors. Explain how doping changes the electrical properties of a semiconductor.
3) State the Gauss law and mentions its Applications.
4) Define (i) dielectric (ii) Polarization of charges.
5) Define electrostatic potential and the formula associated with it.
6) State and explain the depletion region of a p-n junction diode increases with reverse bias and decreases with forward bias.
7) Write a short notes on P-n junction and also draw a proper diagram.

8)(a)

If $\phi = x^2 y z^3 + x y^2 z^2$. Find $\vec{\nabla}\phi$ at (1,3,2)

(b) find the gradient of scalar field $\phi = \sqrt{y^2+z^2}$